



DIGITAL INVESTFAIR · FIN 7870

Pricing an Option

from first principles

Hang Seng Index Options, the mathematics behind the trade

What is an option?

A contract that gives you the right, not the obligation, to trade at a fixed price.

CALL

The right to BUY at the strike K.

You profit when the market rises ▲

PUT

The right to SELL at the strike K.

You profit when the market falls ▼

You pay a premium up front, the price of the choice.

Five words to own

S

Underlying

the spot price right now

K

Strike

the locked-in price

P

Premium

what the option costs

T

Expiry

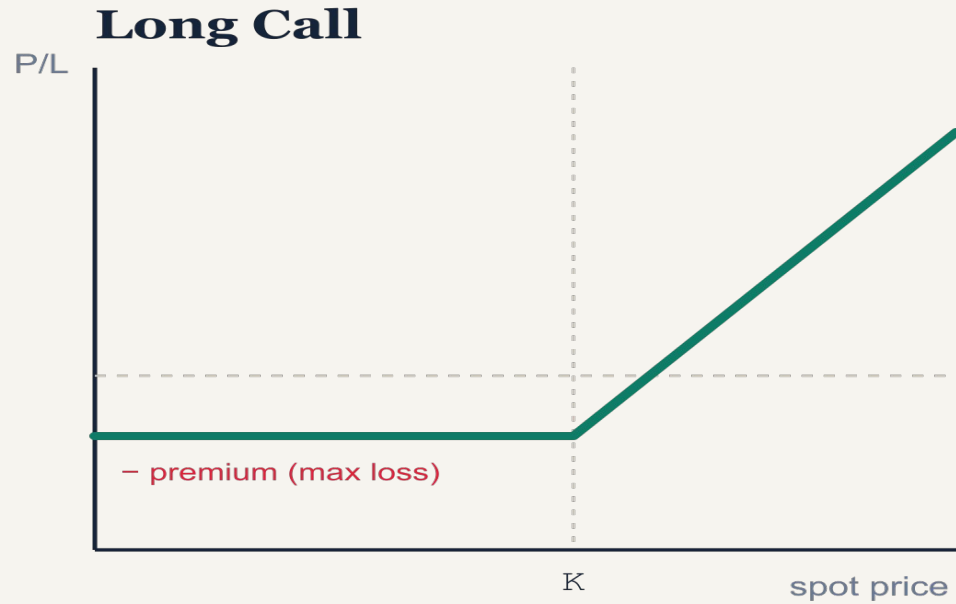
when the choice ends

M

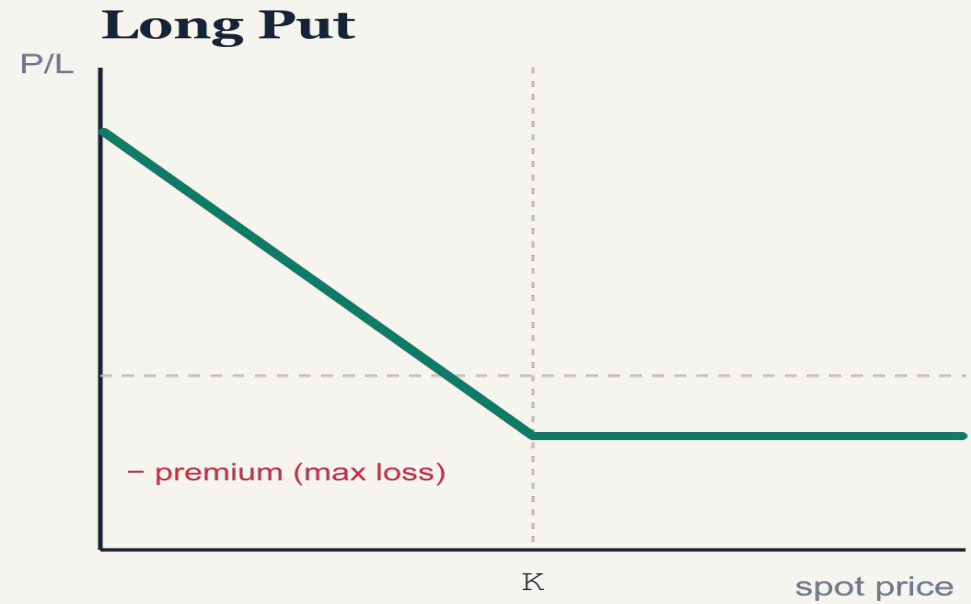
Moneyness

ITM / ATM / OTM

Loss capped, upside open



$$\text{payoff} = \max(S - K, 0)$$



$$\text{payoff} = \max(K - S, 0)$$

A long option can lose only the premium, but its gains are unbounded. That asymmetry is the whole point.

In, at, or out of the money

Call

- ITM** spot above strike ($S > K$)
- ATM** spot at the strike ($S = K$)
- OTM** spot below strike ($S < K$)

Put

- ITM** spot below strike ($S < K$)
- ATM** spot at the strike ($S = K$)
- OTM** spot above strike ($S > K$)

Premium = intrinsic value + time value. Time value pays for what spot might still do, and it fades to zero by expiry.

Six inputs

Input (rises)	Call value	Put value
Spot price S	▲	▼
Strike K	▼	▲
Time to expiry T	▲	▲
Volatility σ	▲	▲
Interest rate r	▲	▼
Dividend q	▼	▲

Volatility is the big one.

It is the only input you cannot read off a screen, and it drives the premium most.

These sensitivities are the Greeks.

Five sensitivities

Δ

Delta

value vs. spot price

Γ

Gamma

how fast Delta moves

Θ

Theta

value lost to time
decay

v

Vega

value vs. volatility

ρ

Rho

value vs. interest
rates

Three models, one fair value

Binomial Tree

Step the price up or down, then fold the value back to today.

Best for: American & barrier options. Intuitive, what this project uses.

Black–Scholes

One closed-form equation for the price.

Best for: European vanilla options. Instant, exact.

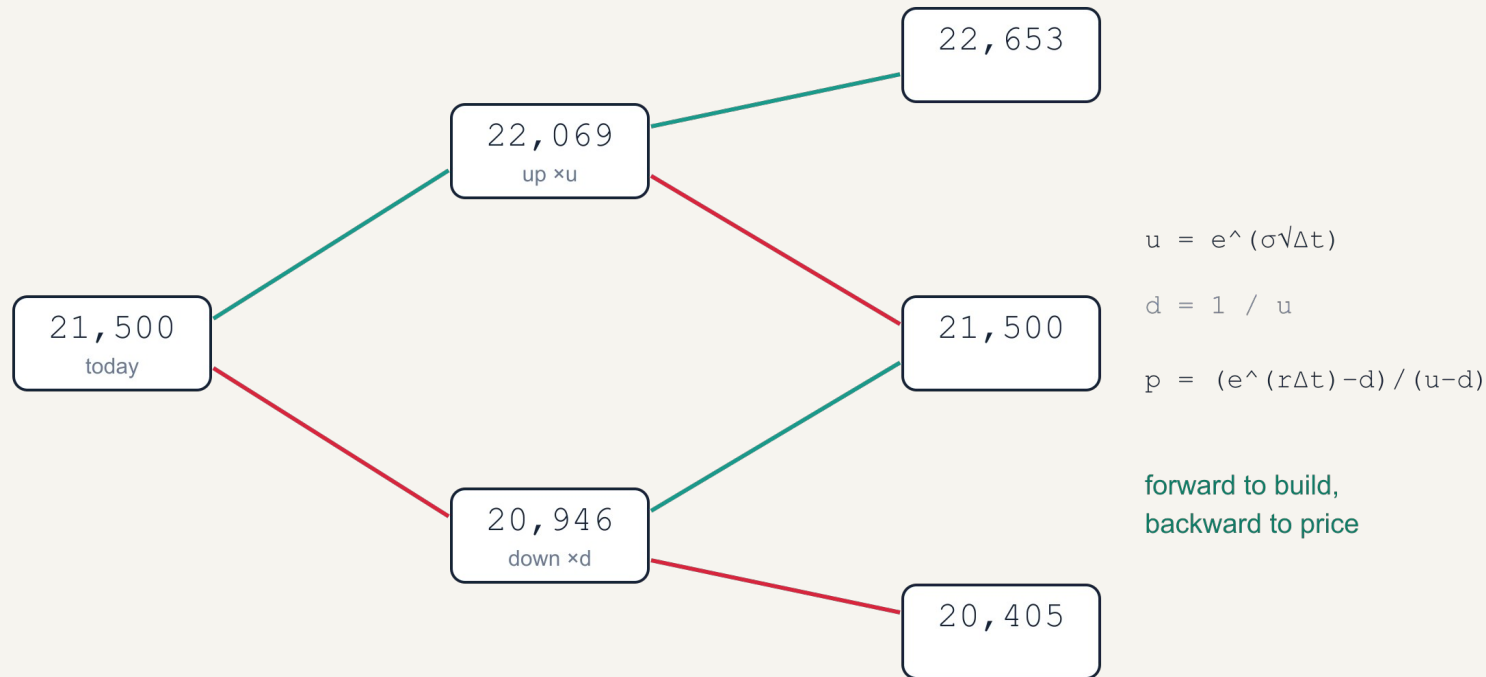
Monte Carlo

Simulate thousands of random paths, then average the payoff.

Best for: complex, path-dependent exotics.

Build forward, price backward

Binomial tree • CRR • 2 steps shown



Backward induction

At expiry the payoff is obvious.
Discount each step by

$$e^{(-r\Delta t)}$$

and fold back to today:

fair value = 186 pts

S spot • u up factor • d down factor • p risk-neutral prob • r rate • σ vol • Δt step • $e^{(-r\Delta t)}$ discount

One node, worked out

1 • Two outcomes

$$S_0 = 21,500$$

$$\text{up} \rightarrow 21,500 \times u = 22,069$$

$$\text{down} \rightarrow 21,500 \times d = 20,946$$

2 • Risk-neutral p

$$p = (e^{(r\Delta t)} - d) / (u - d) \\ = 0.504 \quad 1 - p = 0.496$$

3 • Payoffs at $K = 22,000$

$$C_u = \max(22,069 - 22,000, 0) = 69$$

$$C_d = \max(20,946 - 22,000, 0) = 0$$

4 • Discount one node

$$C = e^{(-r\Delta t)} [p \cdot C_u + (1-p) \cdot C_d] \\ \approx 0.504 \times 69 \approx 34.7$$

One node of the 3-step grid ≈ 34.7 . **Same rule, finer grid: 3 steps $\rightarrow$ fair value 186.**

Risk-neutral valuation

p is a pricing weight, not a probability

$$p = (e^{(r\Delta t)} - d) / (u - d)$$

A pricing weight, not the real chance of a rise.

Real-world drift never enters.

Why we discount at r

Pretend every asset grows at the risk-free rate r .

Then $\text{price} = e^{-rT} \times \text{expected payoff}$.

No risk premium, no risk appetite needed.

The same risk-neutral logic sits under Black-Scholes too.

The continuous limit

$$C = S \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$$

$$d1 = [\ln(S/K) + (r + \sigma^2/2) T] / (\sigma\sqrt{T})$$

$$d2 = d1 - \sigma\sqrt{T}$$

As the tree's steps shrink toward zero, it converges to this single formula. Same fair value, computed instantly.

How the desk stays flat

Cancel the risk

Writing a call leaves you short the market.
Hold Δ units of the index to offset.
Net exposure $\rightarrow 0$: the book is flat.

Re-hedge as Δ drifts

Δ moves when spot moves; that drift is Gamma.
Trim or add the underlying to stay flat.
Each adjustment locks in a small cost.

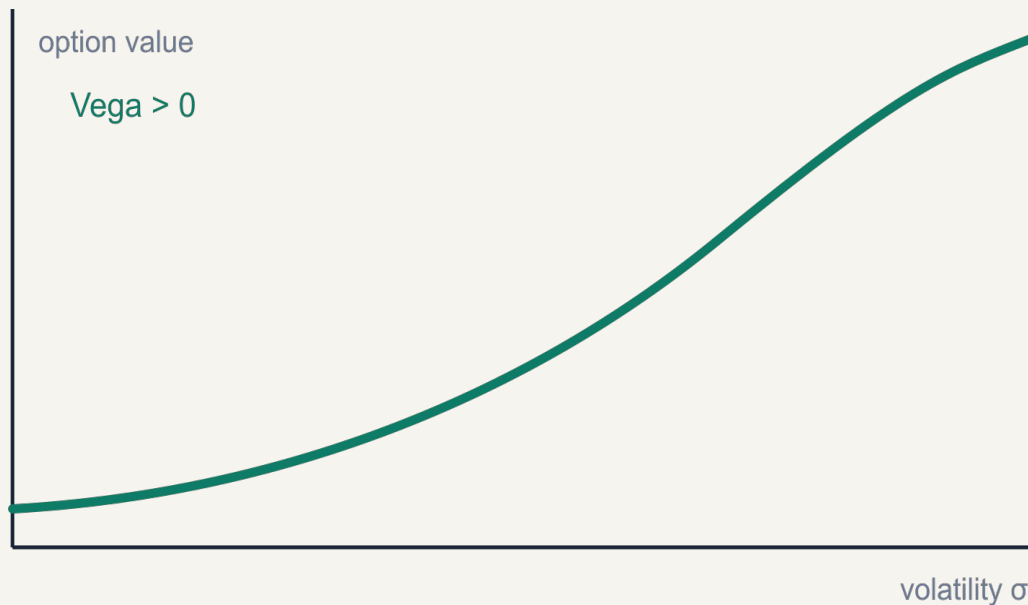
Replication = price

An index + cash portfolio copies the payoff.
Rebalanced each step, it tracks the option.
Its running cost IS the fair value.

Hedge Δ , fund Γ : the running cost of replication is the premium.

The only input you cannot see

Higher volatility → higher value



Spot, strike, time, rate, all observable.

Volatility is the market's estimate of how violently the price will move, its fear. It drives the premium more than anything else.

Solve it backward from prices → implied volatility.

The volatility smile

One vol, all strikes?

Black-Scholes uses one σ for every strike.
Real option prices disagree.
The wings trade richer.

The smile

Plot implied vol against strike.
It rises away from the money.
Both wings curl up: a smile.

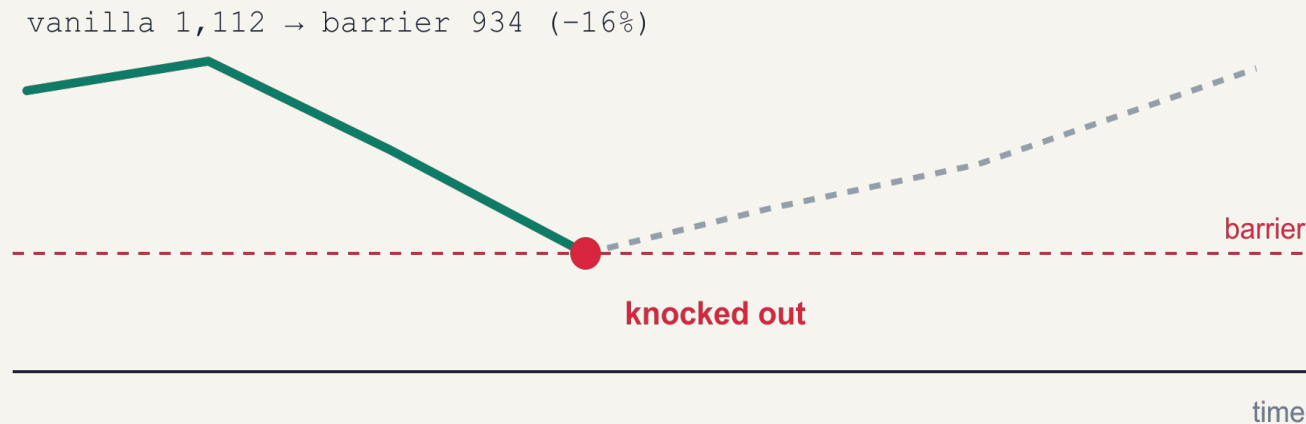
Equity-index skew

Indices tilt, they don't smile evenly.
Crash fear bids up the OTM puts.
Low strikes carry the highest vol.

One model, one σ . Real prices reveal many.

Cheaper, with a tripwire

Down-and-out call · the tripwire



Knock-out

Touch the barrier once and the option dies, worthless, instantly.

**Less protection,
lower premium:
1,112 → 934**

The barrier family

Down-and-out

Dies if spot falls to the barrier.
Barrier below the start (the deck's Day 3 trade).

Down-and-in

Born only if spot falls to the barrier.
Dormant until that level is hit.

Up-and-out

Dies if spot rises to the barrier.
Barrier above the start, usually paired with puts.

Up-and-in

Born only if spot rises to the barrier.
No touch, no option.

Same strike, expiry and barrier: knock-in + knock-out = the vanilla option.

Calls and puts are linked

$$C - P = S - K \cdot e^{-rT}$$

Hold a call and sell a put, and you have replicated the stock itself. Arbitrage forces this to hold, so knowing one option's price gives you the other's.

C call · P put · S spot · K strike · r rate · T time

Hang Seng Index Option

Underlying	Hang Seng Index (HSI)
Contract multiplier	HK\$50 per index point
Exercise style	European, exercised only at expiry
Settlement	Cash (no shares change hands)
Exchange	HKEX · listed since 1993

Asia's flagship equity-index derivative, and the contract priced throughout this project.

■ TRY IT YOURSELF

From theory to the trading desk

We built a game that runs this exact pricing engine: four trading days, real clients to quote, a grade at the end.

Scan to play in your browser:

`zeref007.github.io/trader-simulator`

With thanks to Professor Martin, the inspiration behind this project.



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